

Enhancing Small Language Models in RAG via Triplet-Guided In-Context Distillation

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Abstract

Retrieval-Augmented Generation (RAG) is widely used for knowledge-intensive question answering (QA) but typically relies on large language models (LLMs) as generators, which raises cost and privacy concerns. We propose TripletRAG, a simple, training-free method that converts LLM-teacher behavior into triplets (synthetic question, retrieved passages, reference answer), which are then provided as exemplar context to a small language model (SLM) at inference. TripletRAG requires only offline synthetic QA generation and retrieval; no SLM finetuning is needed. Across SQuAD and Natural Questions (KILT) datasets, triplet guidance substantially increases judge-based accuracy, and improves groundedness on SQuAD for models in the 1B–3B parameter range, with negligible or inconsistent gains for larger models (7B+). Compared to a strong lightweight baseline (MiniRAG), TripletRAG gives markedly better groundedness and correctness while keeping the pipeline simple and practical.

1 Introduction

Retrieval-Augmented Generation (RAG) (Lewis et al., 2020) enables language models (LMs) to answer knowledge-intensive questions (Zhao et al., 2024; Gupta et al., 2024) by conditioning generation on retrieved documents. In practical deployments (Wang et al., 2025), however, open-source small LMs (SLMs) are often preferred for cost, latency, and privacy reasons (Huang et al., 2026). These models remain less reliable at using retrieved evidence, which can lead to ungrounded or incorrect answers. Improving SLM-based RAG is therefore important for systems that must balance answer fidelity, compute, and data control.

We introduce TripletRAG (Fig. 1), a training-free method that transfers LLM-teacher behavior to an SLM-student through **in-context distillation**, i.e., inference-time transfer via prompting while

keeping the student’s parameters frozen. Offline, an LLM produces triplets of (synthetic question, retrieved passages, reference answer). At inference, relevant triplets are supplied as structured in-context examples to the SLM generator, requiring no finetuning or parameter updates.

We experimentally demonstrate that TripletRAG improves answer accuracy and groundedness over vanilla RAG and a lightweight RAG modification (Fan et al., 2026), with dataset-dependent effects on faithfulness. The method is primarily effective for SLMs in the 1B–3B parameter range, while models at 7B and above show diminishing or inconsistent gains.

2 Related Work

The general RAG solutions include Chain-of-Thought prompting (Wei et al., 2022; Feng et al., 2024), ReAct (Yao et al., 2023), and highly useful in practical cases, few-shot prompting (Mueller et al., 2023; Gabouj et al., 2025; Zhao et al., 2025). Enterprise-grade RAG systems are primarily designed for large-scale LMs as backbone models. Hence, there remains a demand for resource-efficient and privacy-preserving solutions.

SLM-based RAG remains challenging because smaller models are weaker at complex query interpretation, multi-step reasoning, semantic matching, and evidence synthesis (Fan et al., 2026). Prior methods address these limitations through additional training or specialized retrieval architectures. RoseRAG (Liu et al., 2025) improves robustness to noisy retrieval using multi-turn prompting, rejection sampling, and contrastive preference data selection, but requires a computationally expensive training pipeline and substantial annotated data. DRAG (Chen et al., 2025) distills LLM-generated evidence and graphs into an SLM, but does not rely on an external database and is limited by the teacher’s internal knowledge. MiniRAG (Fan et al.,

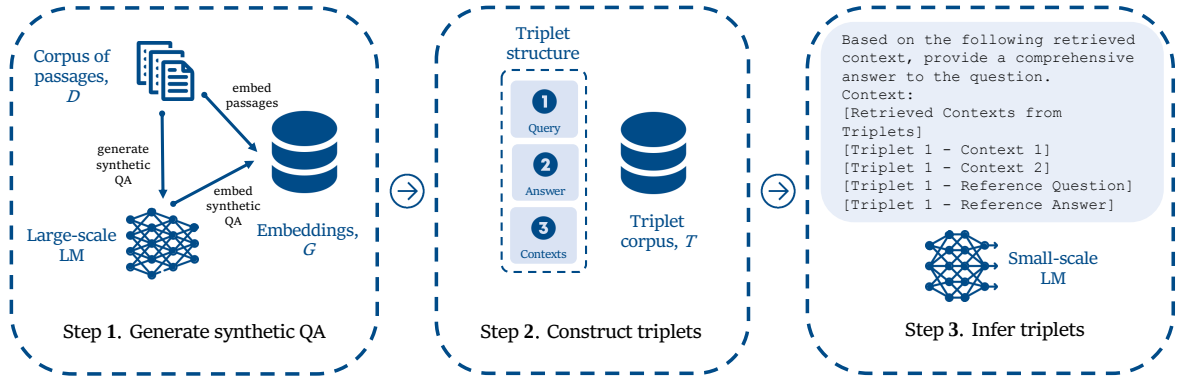


Figure 1: Proposed triplet-guided RAG method in scheme

2026) introduces semantic-aware heterogeneous graph indexing and topology-enhanced retrieval, but still depends on SLM evidence-mining capabilities. In contrast, TripletRAG is training-free and improves generation by providing teacher-derived retrieval examples at inference time.

A related line of work improves retrieval through document expansion and QA-database retrieval. Doc2query and docTTTTTquery (Nogueira and Lin, 2019) augment passages with synthetic queries at indexing time; QUADRo (Campese et al., 2023) answers questions by retrieving similar questions from a QA database; and QuOTE (Neeser et al., 2025), which we use in our pipeline, leverages synthetic QA pairs for dense retrieval. Unlike these methods, TripletRAG uses synthetic QA pairs as structured in-context exemplars at the generation stage, while leaving the retrieval pipeline unchanged. Thus, our contribution is orthogonal: we improve how SLMs consume retrieved evidence, not how evidence is retrieved. We therefore present TripletRAG not as a new retrieval or RAG paradigm, but as a specific empirical recipe for retrieval-augmented few-shot prompting, where synthetic QA exemplars are selected at inference time to demonstrate how retrieved evidence should be used to answer a query.

3 Methods

In this section, we introduce a method designed to enhance SLMs for question-answering (QA) in RAG settings¹. We assume that SLMs can benefit from exposure to the outputs of retrieval-augmented reasoning from larger LMs. Specifically, we use a large LM to generate a synthetic

QA pair set and to retrieve answer supporting contexts offline. The generated set and retrieval are then utilized as structured examples for a smaller LM during inference. Let us describe the details of the proposed approach (Figure 1).

Step 1.1. Generate a synthetic QA pair set.

Prior work has shown that generating a synthetic QA set leads to more accurate retrieval, as proven in this article (Neeser et al., 2025). Thus, the first step in acquiring contextual knowledge from a large LM is generating a new set of queries and answers for all given passages. Given a corpus of passages D , we generate synthetic QA pairs for each passage using a large instruction-tuned language model (see our prompt in Appendix E.1).

Step 1.2. Embed queries and passages.

To retrieve contexts, we use KNN upon query and passage embeddings. The generated QA set and corpus passages are embedded to obtain the embedding set G . For synthetic queries, we embed the concatenated question + answer rather than the question alone; empirically, this yields better retrieval quality (higher context accuracy and MRR, see Appendix C.3), because the answer provides disambiguating signals for which passages are evidence for the intended response.

Step 2. Construct triplets.

For each generated query embedding in G , we retrieve the top N_{ct} passages from a vector store. Each triplet consists of: (i) the generated question, (ii) the retrieved passages, and (iii) the reference answer associated with the question. We denote the set of all triplets as T . The format comprises both a QA example and its retrieval, which is beneficial for a small LM.

Step 3. Infer triplets as a retrieval source.

At inference time, answers are generated for the original benchmark questions. For a given input

¹Source code is publicly available at <https://github.com/ti3c2/triplet-rag-paper>

query, we retrieve a small number of synthetic questions from the embedding set G and obtain the corresponding passage-level triplets from T . To avoid redundancy, synthetic questions associated with the same underlying passage are deduplicated, following the passage-level deduplication strategy introduced in QuOTE (Neeser et al., 2025). The retrieved triplets are then provided to the SLM. The prompt for triplet-based RAG inference is provided in Appendix E.2. TripleRAG introduces no additional passages beyond those in the retrieved triplets; the only prompt overhead is the synthetic Q–A wrapper. Pre-fill latency (p50/p95) is statistically identical across setups: 425/842 ms for Vanilla RAG vs. 424/814 ms for TripletRAG. Measured on the 10K evaluation sets, TripletRAG adds on average 48 input tokens per query on SQuAD (1,701 $\rightarrow$ 1,749) and 40 on NQ (1,288 $\rightarrow$ 1,328), i.e. under 3%. Generated answers are at the same time shorter under triplet guidance (e.g. 109.5 $\rightarrow$ 78.2 output tokens per query for Qwen2.5-7B on SQuAD), so the combined input–output budget of the two setups differs by at most about 2%.² Since both setups use the same instruction wording (Appendices E.2 and E.3), this reduction is attributable to the reference answer in the exemplar rather than to prompt phrasing.

We hypothesize two complementary mechanisms by which triplets help SLMs: (1) scaffolding: the synthetic question + reference answer gives the SLM an explicit example of how retrieved passages map to a high-quality answer format; (2) retrieval alignment: including the retrieved passages alongside the synthetic QA encourages the SLM to ground its generation on actual evidence rather than hallucinating. Our ablations (Appendix C) empirically target these mechanisms.

4 Experimental Setup

Experiments were conducted on the established open-domain QA benchmarks SQuAD (Rajpurkar et al., 2016) and the KILT version (Petroni et al., 2021) of Natural Questions (NQ) (Kwiatkowski et al., 2019), containing fully parsed Wikipedia passages including span annotations. The evaluation sample consists of the first 10,000 questions from each dataset. All generative metrics are computed

²Token counts are computed with a common BPE tokenizer rather than the native tokenizer of each backbone; the triplet-side input delta is estimated from the average length of the synthetic Q–A wrapper.

over this full sample and under an identical protocol for Vanilla RAG, TripletRAG and MiniRAG, so that the three systems are directly comparable.

Synthetic QA pairs are generated using models Qwen2.5-32B-Instruct, Llama-3.1-70B-Instruct (Yang et al., 2025; Grattafiori et al., 2024) with a temperature of 0.7. Structured output constraints are applied to ensure valid formatting. All passages and synthetic QA pairs are embedded using jina-embeddings-v3 (Sturua et al., 2025). When constructing triplets, each synthetic QA pair is connected to $N_{ct} = 5$ retrieved passages. Then, at the inference phase, $N_{qt} = 2$ nearest synthetic QAs are retrieved for each input query, resulting in $N_e = 10$ retrieved passages. All models used to answer questions during RAG inference are evaluated with a decoding temperature of 0.1.

The method is compared against a standard baseline RAG setup (vanilla RAG) and MiniRAG (Fan et al., 2026). We compute standard retrieval metrics following the QuOTE evaluation protocol (Neeser et al., 2025), including *context accuracy* and *title accuracy*. Generative performance is evaluated using the RAGAS framework (Es et al., 2024), reporting *faithfulness*, *answer accuracy*, and *groundedness*. As a judge model, we select Qwen2.5-32B-AWQ with a temperature of 0.01. Importantly, when evaluating context usage, we provide the judge model only with the retrieved passages for each triplet, excluding the associated synthetic questions and answers, as these represent the retrieved evidence. Metric definitions and evaluation details are provided in Appendix A. All generative scores in Tables 1 and 2 are averaged over the full 10,000 questions per dataset. The ablation studies in Tables 4 and 5 are run on smaller subsets, which are stated in the corresponding table captions. Following standard QA evaluation practice, we additionally report token-level F1 computed using the SQuAD normalization protocol (lowercasing, removal of articles and punctuation).

5 Experimental Results

Main results for the Qwen2.5 model family (Yang et al., 2025) are summarized in Table 1, while Table 2 in Appendix A.2 contains results for Llama models (Meta AI, 2024; Grattafiori et al., 2024).

Retrieval Metrics. We observe a slight decrease in retrieval performance when using triplet-guided RAG. On SQuAD, context accuracy decreases

Setup	Backbone model	CAcc.	TAcc.	Faith.	NV Acc.	Ground.	F1
		SQuAD					
Vanilla RAG	Qwen2.5-7B	93.69	98.79	84.61	70.51	92.97	6.60
	Qwen2.5-3B	93.69	98.79	79.42	58.83	85.23	5.69
TripletRAG (ours)	Qwen2.5-7B	91.69	98.41	84.46	70.55	94.28	9.92
	Qwen2.5-3B	91.69	98.41	84.71	70.11	94.06	12.20
MiniRAG (Fan et al., 2026)	Qwen2.5-7B	–	–	75.61	43.79	89.77	3.54
	Qwen2.5-3B	–	–	80.51	42.15	88.28	5.24
		Natural Questions					
Vanilla RAG	Qwen2.5-7B	82.84	85.07	86.44	44.36	78.44	5.65
	Qwen2.5-3B	82.84	85.07	83.03	38.75	73.92	5.87
TripletRAG (ours)	Qwen2.5-7B	77.83	80.22	84.82	44.38	77.23	6.89
	Qwen2.5-3B	77.83	80.22	80.35	43.28	76.05	7.40
MiniRAG (Fan et al., 2026)	Qwen2.5-7B	–	–	63.01	38.82	76.15	2.83
	Qwen2.5-3B	–	–	71.51	34.02	74.05	4.45

Table 1: RAG metrics comparison. Questions for TripletRAG are generated with Qwen2.5-32B-Instruct. CAcc. – Context Accuracy, TAcc. – Title Accuracy, Faith. – Faithfulness, NV Acc. – Accuracy (judge-based correctness), Ground. – Groundedness (evidence use), F1 – token-level F1. All generative metrics are computed with Qwen2.5-32B-AWQ as judge over all 10,000 evaluation questions per dataset.

slightly (by approximately 1–2 points), while title accuracy remains essentially unchanged. On NQ, both context and title accuracy decrease by approximately 5–8 points. This indicates that improvements in answer grounding are not driven by higher retrieval accuracy.

Faithfulness Results. We analyze the effect of triplet-guided RAG on answer faithfulness, measuring the extent to which generated answers are supported by retrieved evidence. The effect of triplet guidance is capacity- and dataset-dependent. For Qwen2.5-7B-Instruct, faithfulness remains essentially unchanged on SQuAD (84.61 → 84.46) and decreases on NQ (86.44 → 84.82). For Qwen2.5-3B-Instruct, faithfulness increases on SQuAD (79.42 → 84.71) but decreases on NQ (83.03 → 80.35). Thus, the clearest faithfulness gain is observed for the 3B model on SQuAD, while the effect is not consistent across datasets.

NV Accuracy Results. We additionally analyze NV Accuracy, which measures agreement between generated answers and reference answers using an LLM-based judge. For the 7B model, NV Accuracy remains essentially unchanged under TripletRAG on both SQuAD (70.51 → 70.55) and NQ (44.36 → 44.38). For the 3B model, TripletRAG yields a substantial increase on SQuAD (58.83 → 70.11) and a more moderate increase on NQ (38.75 → 43.28). Thus, the NV Accuracy gains are concentrated in the smaller model rather than being uniform across model sizes. A query-similarity ablation (Table 6) suggests this improvement is not explained by merely retrieving near-

duplicate synthetic questions and memorizing their answers, but reflects better evidence use for the target query. Token-level F1 corroborates these findings: on SQuAD, TripletRAG improves F1 from 6.60/5.69 (Vanilla RAG, 7B/3B) to 9.92/12.20, while MiniRAG scores 3.54/5.24. On NQ, gains are more modest (5.65/5.87 → 6.89/7.40). These NV Accuracy gains are reproduced by an independent judge from a different model family (Appendix A.4; +12.30 for 3B on SQuAD and +4.50 on NQ, versus no gain for 7B).

Groundedness Results. We next examine the impact of triplet-guided RAG on groundedness, a measure of whether the retrieved context supports the generated answers. On SQuAD, groundedness improves for Qwen models at both 7B and 3B scales, increasing from 92.97 → 94.28 for Qwen2.5-7B-Instruct and from 85.23 → 94.06 for Qwen2.5-3B-Instruct. On NQ, the effect depends on model size: groundedness decreases slightly for 7B (78.44 → 77.23) but increases for 3B (73.92 → 76.05).

Comparison with MiniRAG. In Table 1, we also compare our approach against strong lightweight baseline, MiniRAG (Fan et al., 2026) (see details in Appendix A.1).

Crucially, our TripletRAG substantially improves judge-based correctness (NV Accuracy) over MiniRAG by approximately 27–28 points and groundedness by 4.5–5.8 points on SQuAD, and remains training-free and simple to deploy.

Ablation Results. Removing the synthetic QA fields while keeping the same retrieved pas-

sages yields small, capacity-dependent effects, with smaller models (3B) benefiting more from explicit QA scaffolding (Appendix C.1). Substituting semantically irrelevant triplets consistently degrades all generative metrics, confirming that gains require semantic alignment rather than a formatting effect (Appendix C.2). Using synthetic QA without retrieved passages leads to a dramatic drop in faithfulness and groundedness, underscoring that retrieved evidence is essential.

6 Conclusion

We introduced TripletRAG, an inference-only method that leverages LLMs to create triplets (synthetic question, retrieved passages, reference answer) and uses them as structured in-context guidance for small LMs in RAG pipelines. Without any SLM finetuning, TripletRAG significantly improves judge-based correctness on SQuAD and NQ and increases groundedness on SQuAD, with the clearest benefits for models in the 1B–3B parameter range and diminishing returns for larger models (7B+). Our ablations show that gains depend on triplet semantic relevance rather than formatting effects. TripletRAG is practical for deployed settings because it decouples expensive LLM reasoning (done offline) from lightweight inference. Evaluation on more challenging datasets is left for future work.

Limitations

Triplet construction in the proposed approach requires LLM inference and offline retrieval, which may be costly for very large corpora, even if it can be parallelized and batched. TripletRAG also inherits toxicity and hallucination risks of the LLM-generated answers. Finally, the benefits of TripletRAG are concentrated in lower-capacity SLMs and are dataset-dependent: gains are clearest on SQuAD, more modest on NQ, and diminish or disappear at 7B and above. We therefore view TripletRAG as a practical option primarily for cost-sensitive or privacy-constrained deployments.

Our main evaluation relies on a judge from the same model family as one of the triplet generators. A check with an independent judge (Appendix A.4) reproduces the judge-based accuracy gains but does not reproduce the groundedness gains on NQ; conclusions about groundedness should therefore be read as judge-dependent outside SQuAD.

We do not compare against a dynamic few-shot

baseline in which real labelled QA pairs are retrieved by similarity and used as exemplars. Such a baseline assumes access to annotated question-answer data, whereas TripletRAG requires only an unlabelled passage corpus and constructs its exemplars offline; we therefore regard it as a stronger-supervision reference point rather than a matched baseline. Isolating the contribution of synthetic triplets against retrieved real exemplars remains open.

Ethical Considerations

The experiments reported in this paper were conducted objectively and fairly. No biases were introduced during the data selection or evaluation process. All datasets used in this research are publicly available, and the methods were rigorously tested to ensure the reliability and validity of the results.

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Appendix A Complete Evaluation Results

This section presents the primary quantitative results for the proposed RAG method and baseline systems discussed in the main text, as shown in Table 2. Results are grouped by dataset and generator model size, and include both retrieval and generative evaluation metrics.

In the *Setup* column, *Vanilla RAG* denotes the standard RAG pipeline that retrieves passages whose embeddings are closest to the input query embedding, while *TripletRAG* denotes the proposed triplet-guided RAG method.

A.1 MiniRAG Comparison

To compare our method with MiniRAG (Fan et al., 2026), we ran the official implementation on our evaluation data. By design, MiniRAG operates on two types of artifacts: *Entities* (extracted entities and their definitions) and *Contexts* (chunks of text extracted by the large LM), which together serve as the model’s input context. For knowledge-graph construction and other preprocessing steps, we use Qwen2.5-32B-Instruct, while smaller models are used for RAG inference. For RAGAS evaluation, we include both artifacts as retrieved contexts for the judge model, so evidence-based metrics (e.g., faithfulness and groundedness) are computed using the full information MiniRAG provides at

inference time. We do not report Context Accuracy and Title Accuracy for MiniRAG, since its evidence is generated rather than retrieved from the original passage corpus.

A.2 Additional Results: Llama Models

Faithfulness. For Llama-family generators, the effect of triplet guidance depends on both model size and dataset. On SQuAD, faithfulness improves for Llama-3.2-1B (76.67 $\rightarrow$ 83.74–84.38), while Llama-3.2-3B decreases (89.68 $\rightarrow$ 79.99–81.85). On NQ, we observe the same tendency: faithfulness decreases for Llama-3.2-3B (85.79 $\rightarrow$ 80.20), while Llama-3.2-1B shows a slight increase (63.59 $\rightarrow$ 64.52).

NV Accuracy In Vanilla RAG, Llama-3.2-3B has higher NV Accuracy than Qwen2.5-3B on SQuAD (63.36 vs. 58.83) and the same NV Accuracy on NQ (38.75). Under triplet guidance, NV Accuracy improves substantially for Llama-3.2-1B on both datasets (51.65 $\rightarrow$ 63.41–63.70 on SQuAD and 33.89 $\rightarrow$ 38.87 on NQ). For Llama-3.2-3B, the improvement is modest on SQuAD (63.36 $\rightarrow$ 64.08–65.19), while NV Accuracy decreases on NQ (38.75 $\rightarrow$ 34.87).

Groundedness. We observe consistent decreases in groundedness for Llama-3.2-3B across both datasets. In contrast, the smallest model, Llama-3.2-1B, benefits substantially from triplet guidance, with groundedness improving from 80.84 $\rightarrow$ 88.54–88.97 on SQuAD and from 59.15 $\rightarrow$ 64.71 on NQ (Table 2).

A.3 Cross-Generation Experiments

We also evaluate cross-family triplet generation to test whether triplet-guided RAG depends on using the same model family for synthetic QA construction and downstream RAG. Specifically, we (i) generate triplets with Llama-3.1-70B-Instruct and perform RAG with Qwen2.5-7B-Instruct and Qwen2.5-3B-Instruct, and (ii) generate triplets with Qwen2.5-32B-Instruct and perform RAG with smaller Llama generators. On SQuAD, where both triplet generators are reported for the same downstream backbones, the resulting generative metrics vary only slightly, usually by around 1 percentage point or less, with the largest difference being 1.86 points in faithfulness for Llama-3.2-3B. This suggests that the proposed triplet guidance has limited sensitivity to the triplet-generator model

Setup	QGen LLM	Backbone model	CAcc.	TAcc.	Faith.	NV Acc.	Ground.
			Squad				
Vanilla RAG	–	Qwen2.5-7B	93.69	98.79	84.61	70.51	92.97
		Qwen2.5-3B	93.69	98.79	79.42	58.83	85.23
		Llama-3.2-3B	93.69	98.79	89.68	63.36	92.62
		Llama-3.2-1B	93.69	98.79	76.67	51.65	80.84
TripletRAG	Qwen2.5-32B	Qwen2.5-7B	91.69	98.41	84.46	70.55	94.28
		Qwen2.5-3B	91.69	98.41	84.71	70.11	94.06
		Llama-3.2-3B	91.69	98.41	81.85	64.08	91.80
		Llama-3.2-1B	91.69	98.41	83.74	63.41	88.97
	Llama-3.1-70B	Qwen2.5-7B	92.21	98.80	83.92	71.68	94.49
		Qwen2.5-3B	92.21	98.80	83.88	71.00	94.20
		Llama-3.2-3B	92.21	98.80	79.99	65.19	91.51
		Llama-3.2-1B	92.21	98.80	84.38	63.70	88.54
MiniRAG	–	Qwen2.5-7B	–	–	75.61	43.79	89.77
		Qwen2.5-3B	–	–	80.51	42.15	88.28
			Natural-questions				
Vanilla RAG	–	Qwen2.5-7B	82.84	85.07	86.44	44.36	78.44
		Qwen2.5-3B	82.84	85.07	83.03	38.75	73.92
		Llama-3.2-3B	82.84	85.07	85.79	38.75	74.86
		Llama-3.2-1B	82.84	85.07	63.59	33.89	59.15
TripletRAG	Qwen2.5-32B	Qwen2.5-7B	77.83	80.22	84.82	44.38	77.23
		Qwen2.5-3B	77.83	80.22	80.35	43.28	76.05
	Llama-3.1-70B	Llama-3.2-3B	74.62	77.17	80.20	34.87	71.05
		Llama-3.2-1B	74.62	77.17	64.52	38.87	64.71
MiniRAG	-	Qwen2.5-7B	–	–	63.01	38.82	76.15
	-	Qwen2.5-3B	–	–	71.51	34.02	74.05

Table 2: Effect of proposed triplet-guided RAG. All generative metrics are computed with Qwen2.5-32B-AWQ as judge over all 10,000 evaluation questions per dataset.

family rather than relying strongly on family-specific artifacts.

A.4 Independent-Judge Evaluation

The judge used in our main experiments (Qwen2.5-32B-AWQ) belongs to the same model family as one of the triplet generators and as two of the evaluated backbones. To test whether the reported effects depend on this choice, we re-evaluated all Qwen configurations with gemma-3-27b-it as an independent judge, holding the generated answers, the evaluation questions, and the metric definitions fixed. Results are reported in Table 3.

Absolute scores are not comparable across judges: Gemma assigns systematically higher values than the Qwen judge, most visibly for the Vanilla RAG baseline on SQuAD (81.17 vs. 70.51 NV Accuracy). The relevant quantity is therefore the change induced by triplet guidance.

For NV Accuracy, the two judges agree closely. On SQuAD, TripletRAG improves the 3B model by

11.28 points under the Qwen judge and by 12.30 points under Gemma, while the 7B model is essentially unchanged under both (+0.04 and +1.16). On NQ the agreement is closer still: +4.53 versus +4.50 for the 3B model, and no improvement for 7B under either judge. The capacity-dependence of the judge-based accuracy gains is thus reproduced by an independent judge.

For groundedness the two judges diverge. Under the Qwen judge, triplet guidance improves groundedness in three of four configurations; under Gemma it does so only for the 3B model on SQuAD (90.46 $\rightarrow$ 95.73), while the remaining configurations are flat or negative, most clearly for the 7B model on NQ (87.61 $\rightarrow$ 83.44). We therefore regard the judge-based accuracy gains as robust to the choice of judge, and the groundedness gains as supported on SQuAD but not established on NQ.

Setup	Backbone model	Faith.	NV Acc.	Ground.
SQuAD				
Vanilla RAG	Qwen2.5-7B	92.89	81.17	96.35
	Qwen2.5-3B	84.60	69.74	90.46
TripletRAG (ours)	Qwen2.5-7B	89.81	82.33	96.06
	Qwen2.5-3B	89.46	82.04	95.73
Natural Questions				
Vanilla RAG	Qwen2.5-7B	89.81	53.87	87.61
	Qwen2.5-3B	83.84	47.90	86.04
TripletRAG (ours)	Qwen2.5-7B	87.47	52.78	83.44
	Qwen2.5-3B	84.06	52.40	83.63

Table 3: Independent-judge evaluation with gemma-3-27b-it. Triplets are generated with Qwen2.5-32B-Instruct. All values are computed over the same 10,000 evaluation questions per dataset as Table 1.

Appendix B Metrics

Retrieval Metrics. We report the following retrieval metrics, following the QuOTE evaluation protocol:

- **Context Accuracy (CAcc.):** the proportion of evaluation queries for which the reference passage appears among the retrieved passages.
- **Title Accuracy (TAcc.):** the proportion of evaluation queries for which at least one retrieved passage originates from the correct source document (matched by Wikipedia title).

Generative Metrics. We adopt the RAGAS evaluation framework (Es et al., 2024), which implements LLM-as-a-judge scoring. The following metrics are reported:

- **Faithfulness (Faith.):** fraction of answer claims supported by the retrieved context.
- **Accuracy (NV Acc.):** agreement with the reference answer based on two independent NVIDIA LLM-judge ratings, averaged and normalized to $[0, 1]$.
- **Response Groundedness (Ground.):** degree to which the generated answer is grounded in the supplied context.

The NVIDIA metrics (Accuracy, Response Groundedness) follow the evaluation rubric described in the NeMo Agent Toolkit documentation (NVIDIA, 2025).

Appendix C Ablation Details

This appendix provides additional experimental details and quantitative results for the ablation studies referenced in the main text. We verify that gains are not a mere few-shot formatting artifact: substituting semantically irrelevant triplets degrades groundedness and NV Accuracy (Table 5). Likewise, removing retrieved passages while keeping synthetic Q-A examples leads to dramatic drops (Table 4), confirming that triplets work by coupling exemplar QA with real retrieved evidence. These checks show that TripletRAG depends on semantic relevance and evidence grounding rather than on prompt formatting alone.

C.1 Effect of Question-Answer Conditioning

To assess whether improvements arise from the inclusion of QA information rather than from the retrieved passages alone, we compare the full triplet-based setup against a variant in which only the retrieved passages are provided to the generator. In this setting, the same passages are retrieved as in the full model, but the associated synthetic questions and answers are omitted from the context. This experiment isolates the contribution of explicit QA conditioning beyond passage-level evidence. Table 4 summarizes the corresponding generative metrics. Retrieval metrics (context accuracy 91.96%, title accuracy 98.41%) remain unchanged across configurations and are therefore reported once here.

The effect of QA conditioning depends on model capacity. For the smaller Qwen2.5-3B-Instruct, removing the triplet QA fields leads to a clearer degradation in generation quality, with groundedness dropping from 93.14 $\rightarrow$ 92.59 and NV Accu-

Setup	QGen LLM	Backbone model	Faith.	NV Acc.	Ground.
			Squad		
TripletRAG (ours)	Qwen2.5-32B	Qwen2.5-7B	85.35	69.18	93.16
		Qwen2.5-3B	85.20	68.52	93.14
TripletRAG chunks		Qwen2.5-7B	83.44	70.73	93.38
		Qwen2.5-3B	85.13	66.16	92.59
Only QA		Qwen2.5-7B	29.45	50.47	0.00
		Qwen2.5-3B	21.57	52.35	0.00

Table 4: Effect of question-answer conditioning on generative metrics. In the Only QA setting no retrieved passages are supplied, so response groundedness has no context to be measured against and is reported as 0.00. Metrics are computed over a 3,000-question subsample of the SQuAD evaluation set and are therefore not directly comparable to Table 1.

Setup	Backbone model	Faith.	NV Acc.	Ground.
		Squad		
Vanilla RAG	Qwen2.5-7B	85.65	65.51	93.80
	Qwen2.5-3B	85.08	62.43	92.74
Irrelevant triplets	Qwen2.5-7B	83.92	61.52	90.22
	Qwen2.5-3B	81.26	56.61	84.25
Relevant triplets (ours)	Qwen2.5-7B	85.22	68.15	94.42
	Qwen2.5-3B	85.14	67.67	93.73

Table 5: Effect of triplet relevance on generative performance. Metrics are computed on the passage-disjoint evaluation split described in Appendix C.2 (4,260 questions) and are therefore not directly comparable to Table 1. In this ablation each query is paired with a single triplet ($N_{qt} = 1$) and $N_c = 5$ directly retrieved passages, rather than the $N_{qt} = 2 / N_e = 10$ configuration used in Table 1.

racy decreasing from $68.52 \rightarrow 66.16$, while faithfulness is essentially unchanged ($85.2 \rightarrow 85.1$). In contrast, the larger Qwen2.5-7B-Instruct is comparatively insensitive to this structure: groundedness and NV Accuracy shift slightly upward ($93.16 \rightarrow 93.38$ and $69.18 \rightarrow 70.73$), whereas faithfulness decreases ($85.3 \rightarrow 83.4$). Overall, these deltas are small, but they suggest that explicit triplet-style conditioning is more beneficial for smaller generators, while larger models show little improvement and exhibit mixed effects. In contrast, using the synthetic QA pairs *without* the supporting retrieved passages (Only Q-A) leads to a dramatic drop in faithfulness and groundedness, and a noticeable decrease in NV Accuracy, highlighting that the retrieved evidence is essential and QA pairs alone are not a substitute for grounding.

C.2 Effect of Triplet Relevance

Further examination is focused on whether performance gains are driven by the semantic relevance of retrieved triplets or by a few-shot prompting effect induced by the triplet structure itself. For this experiment, we split the original SQuAD evaluation set of 10K questions into two equal splits

by passages, so that passages used to construct triplets do not appear in the evaluation queries. Triplets are constructed using questions and passages from one split, while evaluation is performed on questions from the other split. For each evaluation query, we retrieve a single triplet ($N_{qt} = 1$) from the triplet set; since this triplet is passage-disjoint from the query, its associated contexts are semantically irrelevant. To ensure the presence of relevant evidence, we additionally retrieve $N_c = 5$ passages directly from the dataset using the evaluation query. During generative evaluation, only these query-relevant passages are provided to the judge model, while the irrelevant triplet contexts are excluded. This setup enables us to distinguish between gains attributable to relevant evidence retrieval and those arising solely from exposure to structured QA examples. Table 5 reports the resulting generative metrics. Retrieval metrics (context accuracy: 86.88%, title accuracy: 98.08%) remain unchanged across configurations and are therefore reported here once.

We split the original SQuAD evaluation set of 10K questions into two approximately equal

Dataset	QGen LLM	EM	UniF1	BiF1	Mean LLM-score	Duplicate part LLM
SQ	Qwen2.5-32B	0.0058	0.6325	0.3966	3.3610	0.0397
	Llama-3.1-70B	0.0112	0.6547	0.4238	3.3482	0.0289
NQ	Qwen2.5-32B	0.0021	0.4548	0.1791	3.2090	0.0210
	Llama-3.1-70B	0.0000	0.4621	0.1857	3.2351	0.0246

Table 6: Semantic similarity of subset original and generated queries.

passage-level splits to ensure that passages used for triplet construction do not appear in evaluation queries. The triplet construction split contains 934 passages corresponding to 5,740 questions, while the evaluation split contains 933 passages corresponding to 4,260 questions. Due to multiple questions per passage, an imbalance in the number of questions between the two splits is expected.

Using semantically irrelevant triplets as one-shot RAG examples results in consistent degradation across all generative metrics, with larger drops for the smaller model. Faithfulness decreases from $85.6 \rightarrow 83.9$ for Qwen2.5-7B-Instruct and from $85.0 \rightarrow 81.3$ for Qwen2.5-3B-Instruct. Groundedness drops more markedly, from $93.80 \rightarrow 90.22$ for Qwen2.5-7B-Instruct and from $92.74 \rightarrow 84.25$ for Qwen2.5-3B-Instruct. NV Accuracy also declines, from $65.51 \rightarrow 61.52$ for Qwen2.5-7B-Instruct and from $62.43 \rightarrow 56.61$ for Qwen2.5-3B-Instruct.

In contrast, TripletRAG with semantically relevant triplets (ours) not only avoids the degradation but improves generation quality, most clearly in NV Accuracy and groundedness. For Qwen2.5-7B-Instruct, Faithfulness/NV Accuracy/Groundedness reach $85.22/68.15/94.42$, and for Qwen2.5-3B-Instruct $85.14/67.67/93.73$. Compared to both Vanilla RAG and the irrelevant-triplet setting, relevant triplets yield consistently higher NV Accuracy and groundedness, with the largest gains observed for the smaller model.

Additional ablations concerned analysing the semantic similarity between the original and generated queries in the dataset that are related to one information passage. To quantify similarity, we computed Exact Match (EM), unigram and bigram token-level F1 scores, and an LLM-based similarity score (with Qwen2.5-32B-Instruct as the LLM judge, scoring similarity from 0 to 4, with scores above 3 treated as duplicates). All metrics except EM were calculated on the set of queries shown a cosine similarity above 0.78 (embedder model

jina-embeddings-v3).

As shown in the Table 6, exact matches are extremely rare ($\leq 1\%$). While semantic similarity is non-trivial - as expected given domain overlap - it is far from near-duplicate behavior. These results indicate that the method operates under semantic generalization rather than retrieval-based caching of identical questions and answers.

Together, these results reinforce that the gains from triplet-guided RAG are not attributable solely to the triplet structure, but rather require semantic relevance between the retrieved triplets and the evaluation query.

C.3 Vectorization Strategy Ablation

We evaluate alternative vectorization strategies for synthetic queries, including (i) embedding the question alone and (ii) embedding the concatenated QA pair. Figure 2 reports Context Accuracy, Title Accuracy, and MRR across $k \in \{1, 5, 20\}$ for two embedding models (jina-embeddings-v3 and all-MiniLM-L6-v2), averaged over LLMs (Qwen2.5-7B-Instruct, llama-3-8b-Instruct). Results indicate that embedding the QA pair yields superior performance across all metrics.

Appendix D Scientific Artifacts and Responsible Use

This work uses publicly available scientific artifacts and creates synthetic QA/triplet artifacts as part of the proposed TripletRAG pipeline. The artifacts are used for research evaluation of retrieval-augmented question answering and are not used to train or fine-tune the evaluated SLM generators.

Datasets. We evaluate on SQuAD (Rajpurkar et al., 2016) and the KILT version of Natural Questions (Petroni et al., 2021). Both are English, Wikipedia-grounded question-answering resources. We use the first 10,000 questions from each dataset for evaluation. SQuAD is distributed under CC BY-SA 4.0, and Natural Questions is distributed under

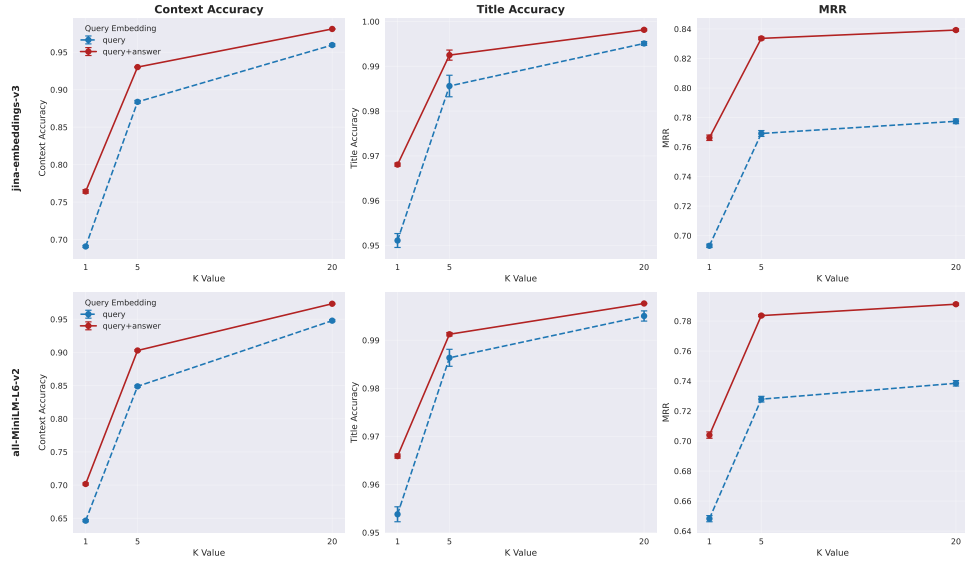


Figure 2: Performance vs. K-Value (averaged over LLMs).

a Creative Commons Share-Alike license. KILT provides a unified benchmark format grounded in a fixed Wikipedia snapshot.

Models and tools. Synthetic QA generation and RAG inference use open-weight Qwen2.5-family models (Yang et al., 2025) and Llama-family models (Grattafiori et al., 2024; Meta AI, 2024). We use jina-embeddings-v3 (Sturua et al., 2025) for embeddings and sentence-transformers/all-MiniLM-L6-v2 (Sentence Transformers, 2021) for the embedding ablation. Generative evaluation uses RAGAS (Es et al., 2024) and NVIDIA NeMo Agent Toolkit metrics (NVIDIA, 2025). For the MiniRAG baseline, we use the official MiniRAG implementation (Fan et al., 2026). The source code for our implementation is released anonymously for review.

Licenses and access terms. We checked the public licenses and access terms of the artifacts used. SQuAD is released under CC BY-SA 4.0. Natural Questions is released under a Creative Commons Share-Alike license. Qwen2.5 models have model-size-dependent licenses: Qwen2.5-7B and Qwen2.5-32B variants are Apache-2.0, while Qwen2.5-3B is governed by the Qwen Research License. Llama 3.1 and Llama 3.2 models are governed by Meta’s Llama Community License. jina-embeddings-v3 is released under CC BY-NC 4.0, sentence-transformers/all-MiniLM-L6-v2 is released under Apache-2.0, MiniRAG is released

under the MIT License, and RAGAS is released under Apache-2.0.

Intended use. Our use is consistent with the intended research and evaluation use of the artifacts. The datasets are used as QA benchmarks; the language models are used for synthetic QA generation, judging, and RAG inference; and the embedding models are used for retrieval and semantic-similarity experiments. The generated synthetic QA pairs and triplets are intended only for research and evaluation of RAG methods. They should not be used as a general-purpose knowledge base or deployed in settings requiring factual guarantees without additional validation.

Privacy, sensitive content, and offensive content.

We do not collect new user data. Natural Questions is described by its creators as consisting of anonymized, aggregated search queries paired with Wikipedia pages. SQuAD and KILT/Natural Questions are Wikipedia-grounded resources and may contain public named entities or other ordinary Wikipedia content. We did not attempt to identify individuals, did not add personal data, and report only aggregate evaluation results. We did not perform additional offensive-content filtering beyond using established public benchmark data; public web- and Wikipedia-derived resources may contain biased, sensitive, or offensive content.

Artifact statistics. For evaluation, we use 10,000 questions from SQuAD and 10,000 questions from Natural Questions. For each passage, the offline

pipeline generates synthetic QA pairs and constructs triplets consisting of a synthetic question, retrieved passages, and a reference answer. In the triplet construction step, each synthetic QA pair is connected to $N_{ct} = 5$ retrieved passages; at inference time, $N_{qt} = 2$ nearest synthetic QAs are retrieved for each input query, resulting in $N_e = 10$ retrieved passages. Additional split statistics for the triplet-relevance ablation are reported in [Appendix C](#).

Appendix E Prompt Templates

This section lists the exact prompt templates used in our pipeline. Specifically, we provide the synthetic QA generation prompt (Section [E.1](#)), the triplet-guided RAG inference prompt (Section [E.2](#)), and the Vanilla RAG inference prompt (Section [E.3](#)) used at evaluation time, for reproducibility.

E.1 Synthetic Question-Answer Generation Prompt

```
System Message:
You are a helpful assistant that
generates questions and answers
for a given text.

User Message:
Generate numerous questions to
properly capture all the
important parts of the
text.

Answer in this json schema:
{
  "items": [
    {
      "question": "<Question 1 text>",
      "answer": "<Answer 1 text>"
    },
    {
      "question": "<Question 2 text>",
      "answer": "<Answer 2 text>"
    }
  ]
}

Text:
...
```

Figure 3: Prompt used to generate synthetic QA pairs from individual passages, following the general structure proposed in the QuOTE framework.

E.2 Triplet RAG Inference Prompt

```
You are a helpful assistant that answers questions based on provided context.

Based on the following retrieved context, provide a comprehensive answer to the
question.
Use only the information from the context and cite relevant parts.
If the context doesn't contain enough information to answer the question,
say so.

Context:

[Retrieved Contexts from Triplets]

[Triplet 1 - Context 1]
...

[Triplet 1 - Context 2]
...

[Triplet 1 - Context 3]
...

[Triplet 1 - Reference Question]
...

[Triplet 1 - Reference Answer]
...

[Triplet 2 - Context 1]
...

[Triplet 2 - Context 2]
...

[Triplet 2 - Context 3]
...

[Triplet 2 - Reference Question]
...

[Triplet 2 - Reference Answer]
...

Question: Which art museum does Notre Dame administer?

Answer:
```

Figure 4: Prompt used for answer generation with retrieved triplets. The example is abbreviated for readability; the actual runs use $N_{ct} = 5$ and $N_{qt} = 2$.

E.3 Vanilla RAG Inference Prompt

```
You are a helpful assistant that answers questions based on provided context.

Based on the following retrieved context, provide a comprehensive answer to the
question.
Use only the information from the context and cite relevant parts.
If the context doesn't contain enough information to answer the question,
say so.

Context:
[Retrieved Context 1]
...
[Retrieved Context 2]
...

Question: Which art museum does Notre Dame administer?

Answer:
```

Figure 5: Prompt used for answer generation in the Vanilla RAG baseline. The example is abbreviated for readability; the actual runs use $N_e = 10$ retrieved passages.